

YAKSHITHA SASANAPURI

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EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY

Integrated M.Tech in Software Engineering (CGPA: 8.73)

Sri Viswa IIT and Medical Academy

Board of Intermediate Education Andhra Pradesh (Grade: 91%)

Ravindra Bharathi School

Board of Secondary Education Andhra Pradesh (Grade: 599/600)

Amaravati, India

Aug 2022 - May 2027

Visakhapatnam, India

August 2020 - May 2022

Pathapatnam, India

May 2020

TECHNICAL SKILLS

Languages: C, Java, JavaScript, Python, SQL

Libraries and Tools: React, PyTorch, Matplotlib, Three.js, MongoDB, Git

Technologies: Visual Studio, Figma, Tailwind CSS, HTML, Shadcn UI, AWS, Claude, Codex, ChatGPT

Skills: DSA, Object-Oriented Programming, REST APIs, Agile Methodology, Web/Mobile Development, CI/CD pipelines

PROFESSIONAL EXPERIENCE

Data Science Intern | Wayspire

- Developed Python-based backend scripts and data pipelines to process and analyze structured datasets, delivering actionable insights to stakeholders.

TECHNICAL PROJECTS

BiteReserve - Campus Canteen Pre-Order & Table Reservation Web App | VIT University

Aug 2025 - Dec 2025

- Designed and developed a campus food-ordering and table reservation web application using React, implementing component-based architecture and reusable UI modules.
- Built end-to-end RESTful data flows for menu selection, cart management, and booking confirmation.
- Implemented responsive, user-friendly interfaces with clear navigation flows, applying OOP design principles for maintainable and scalable frontend code.
- Followed Agile development practices with sprints, code reviews, and continuous integration of new features.
- Wrote unit tests for core booking-logic modules using JavaScript testing utilities, ensuring correctness of reservation conflict detection and cart calculations.

AR Smart Menu – Marker-Based WebAR Food Visualization | VIT University

Dec 2025 - Mar 2026

- Developed a browser-based Augmented Reality application that overlays life-sized 3D food models onto a physical restaurant menu using Mind-AR.js and Three.js.
- Engineered anchor logic for target detection to dynamically show, hide, or persist 3D models (lazy loading).
- Built a content pipeline integrating Generative AI tools for video-to-3D reconstruction, producing optimized .GLB models scaled to real-world dimensions for accurate AR placement.
- Deployed on HTTPS-enabled hosting Vercel to enable mobile camera access via QR code scanning.

Deep Learning-Based Intrusion Detection System | VIT University

Aug 2025 - Dec 2025

- Processed 200k network traffic records and engineered 68 optimized features for IDS.
- Trained CNN, Dense NN, and AutoEncoder models with 99.3 - 99.86% accuracy on CIC-IDS-2017 dataset.
- Built a balanced dataset of 100k samples and deployed 589k-parameter CNN-IDS architecture.
- Implemented end-to-end evaluation using precision, recall, F1-score, confusion matrix, and ROC metrics for robust attack detection performance analysis.

ACHIEVEMENTS

The Institution of Engineers (IEI) | Event Management Lead

Jun 2023 - July 2025

- Headed a cross-functional team of 10 to conceptualize and organize technical competitions for assessing student capabilities.